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CPSC 699 – Assignment 1: Research Result Dissemination

1. General Field of Interest

My general field of interest is Software Engineering. Within software engineering (SE), I am especially interested in software build systems and continuous integration/continuous delivery (CI/CD): build automation, dependency and artifact management, caching, automated testing, pipeline reliability, and developer productivity. This is specific enough to guide my research-venue search while remaining within the broad software-engineering field.

2. Journals in Software Engineering

Below is the list of journals in software engineering with their title, scope and impact factor.

2.1 ACM Transactions on Software Engineering and Methodology (TOSEM):

Impact factor of this journal is 6.9. TOSEM was launched in 1992 and it publishes original, significant, reproducible and archival research across software engineering. Its scope of its published articles includes methods, techniques and tools for specifying, designing, implementing, validating, maintaining and evolving complex, long-lived software systems.

2.2 IEEE Transactions on Software Engineering (TSE):

Impact factor of this journal is 6.0. TSE was established in 1975 and is the oldest journals in Software Engineering. This journal covers core software engineering methodologies, design strategies and development processes. Topics of this journal include development and maintenance methods, testing and validation, reliability, measurement, project management, and software-engineering tools and environments.

2.3 Empirical Software Engineering (ESE):

Impact factor of this journal is 3.4. Established in 1996, ESE publish articles in applied software-engineering research with a strong empirical focus, which includes topics such as empirical evaluation of processes, methods and tools, industrial experience, and quantitative or qualitative evidence about software development and evolution.

3. Reputable Research Conferences in Software Engineering

Below is the list of reputed software engineering conferences with their dates, deadlines and other relevant information.

3.1 International Conference on Software Engineering (ICSE)

ICSE is a premier annual research conference for software-engineering which welcome paper submission in full spectrum of the entire software engineering domain. This includes AI and Foundation model in SE, Architecture and Design, Testing, Repository analytics, DevOps, Human and Social aspect of SE.

URL: https://conf.researchr.org/home/icse-2027

- **Most recent meeting:** The most recent was ICSE 2026 which took place from 12th April to 18th April, 2026 at Rio de Janeiro, Brazil.

- **Next meeting:** The upcoming ICSE 2027 conference will take place from 25th April to 1st May, 2027, in Dublin, Ireland.
- **Proceedings publisher:** ICSE proceedings are usually distributed through the ACM Digital Library and IEEE Xplore. For the upcoming ICSE 2027, the main-track submissions will be using the IEEE conference-proceedings format.
- **Main-event deadline:** For the upcoming ICSE 2027, the deadline for mandatory abstract submission was 23rd June, 2026 and the full research paper submission deadline was 30th June, 2026.
- **Co-located workshops:** ICSE 2027 workshop papers are due on 13th November 2026. The authors will receive acceptance notification on 11th December 2026 with camera-ready submission on 29th January 2027.

3.2 ACM International Conference on the Foundations of Software Engineering (FSE)

FSE is an internationally recognized prestigious conference for researchers, practitioners and educators to present and discuss innovations, trends, experiences and challenges across software engineering. The scope of the conference includes Artificial Intelligence in Software Engineering, System Architecture, Code Analysis, Correctness and Reliability, Empirical and Human Dimensions.

URL: https://conf.researchr.org/home/fse-2027

- **Most recent meeting:** The most recent FSE conference was held from 5th to 9th July 2026 in Montreal, QC, Canada.
- **Next meeting:** The upcoming FSE 2027 will be held in Shenzhen, China from 12th to 16th July 2027.
- **Proceedings publisher:** FSE conference and workshop proceedings are published through the ACM Digital Library.
- **Main-event deadline:** The deadline for the paper submission for FSE 2027 is 2nd October 2026.
- **Co-located workshops:** Workshops for FSE 2027 are scheduled for July 11 and July 17, 2027. Deadline for workshop proposals submission is 15th October 2026 with a hard camera-ready deadline of 8th April 2027.

3.3 IEEE/ACM International Conference on Automated Software Engineering (ASE)

ASE is an annual international conference which focuses on publishing research that improves software development through automation. The conference brings together academic researchers and industry professionals working on methods and tools that support different stages of the software lifecycle. Typical scope of this conference includes AI-assisted development, automated testing and debugging, program analysis, software evolution, mining software repositories, program repair, requirements engineering, and developer-oriented tools.

URL: https://conf.researchr.org/home/ase-2026

- **Most recent meeting:** The most recent ASE 2025 conference was held in Seoul, South Korea from 16th to 20th November 2025.
- **Next meeting:** The upcoming ASE 2026 conference will be held from 12th to 16th October 2026, in Munich, Germany.

- **Proceedings publisher:** ASE is jointly associated with IEEE and ACM. The publication platform and formatting instructions may differ by edition. Authors submitting to ASE 2026 were instructed to prepare their manuscripts using the ACM Primary Article Template
- **Main-event deadline:** Research papers for ASE 2026 had to be submitted by 26th March 2026. Papers invited to undergo major revision had a resubmission deadline of 16th July 2026, while final versions of accepted papers were due on 3rd August 2026.
- **Co-located workshops:** ASE 2026 conference features several workshops and related events scheduled around the main conference, primarily on 12th and 16th October 2026. These workshops do not follow one conference-wide submission timetable.

4. Useful Online Sites and Publications

Some useful websites, newsgroups, and online publications for keeping up with the latest developments in software engineering are listed below:

4.1 ACM SIGSOFT.

The ACM Special Interest Group on Software Engineering (SIGSOFT) serves as a forum for professionals, researchers, and educators in the field of software engineering. It is a central repository for international conference information, institutional policies, prestigious industry awards, community news, and collaborative software engineering initiatives.

4.2 CMU Software Engineering Institute (SEI) Digital Library.

The SEI library contains thousands of reports, papers, presentations and other resources on software engineering, architecture, DevSecOps, cybersecurity and AI engineering.

4.3 InfoQ – Software Engineering / DevOps.

InfoQ is a practitioner-oriented online publication with current articles, news, presentations, podcasts and trend reports in software engineering. It is useful for connecting academic work on CI/CD, architecture and developer productivity with industry practice.

5. Main Industry Players

Software engineering and CI/CD have substantial industry participation. Three major players include Microsoft/GitHub, GitLab, Amazon Web Services (AWS).

5.1 Microsoft / GitHub. GitHub is a widely used platform for source code management and collaborative development. Its CI/CD service, GitHub Actions, automates building, testing, and deployment through event-driven workflows. It also supports dependency caching and artifact management, allowing developers to reuse files and reduce unnecessary build operations.

5.2 GitLab. GitLab provides an integrated DevSecOps platform that combines source code management, CI/CD automation, and security testing. GitLab CI/CD supports automated pipelines, custom runners, dependency caching, and artifact management. Its distributed caching capabilities allow dependencies to be reused across eligible pipeline runs, helping reduce redundant build operations.

5.3 Amazon Web Services (AWS). AWS offers managed CI/CD services for automating software delivery. AWS CodePipeline coordinates the different stages of the delivery process, while AWS CodeBuild handles compilation, testing, and artifact packaging. CodeBuild supports local and Amazon S3-based caching to reduce repeated dependency downloads and improve build efficiency.